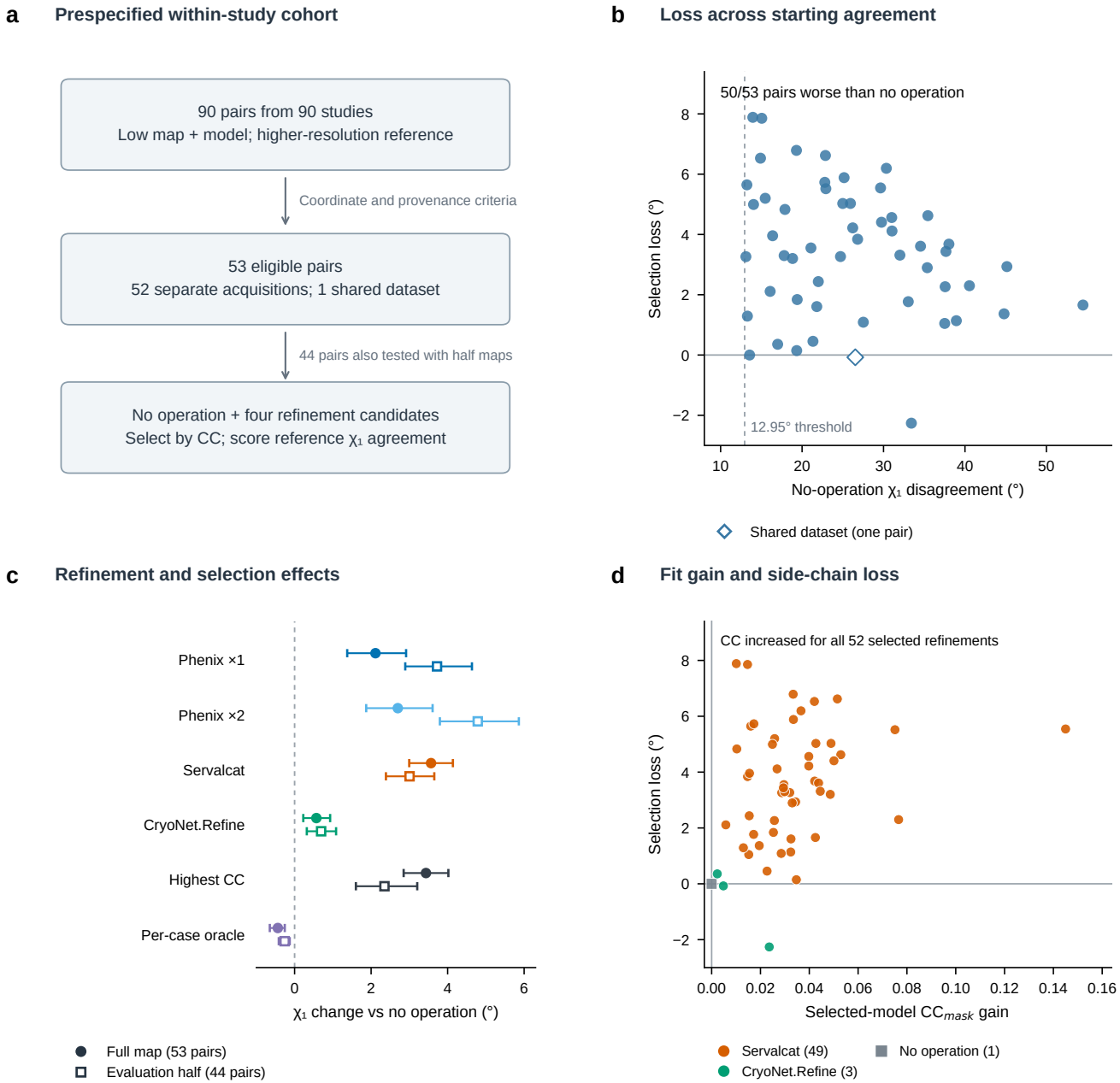
a
Prespecified within-study cohort
90 pairs from 90 studies
Low map + model; higher-resolution reference
Coordinate and provenance criteria
53 eligible pairs
52 separate acquisitions; 1 shared dataset
44 pairs also tested with half maps
No operation + four refinement candidates
Select by CC; score reference χ1 agreement
b
Loss across starting agreement
50/53 pairs worse than no operation
12.95° threshold
Selection loss (°)
No-operation χ1 disagreement (°)
Shared dataset (one pair)
c
Refinement and selection effects
Phenix ×1
Phenix ×2
Servalcat
CryoNet.Refine
Highest CC
Per-case oracle
χ1 change vs no operation (°)
Full map (53 pairs)
Evaluation half (44 pairs)
d
Fit gain and side-chain loss
CC increased for all 52 selected refinements
Selection loss (°)
Selected-model CCmask gain
Servalcat (49)
No operation (1)
CryoNet.Refine (3)